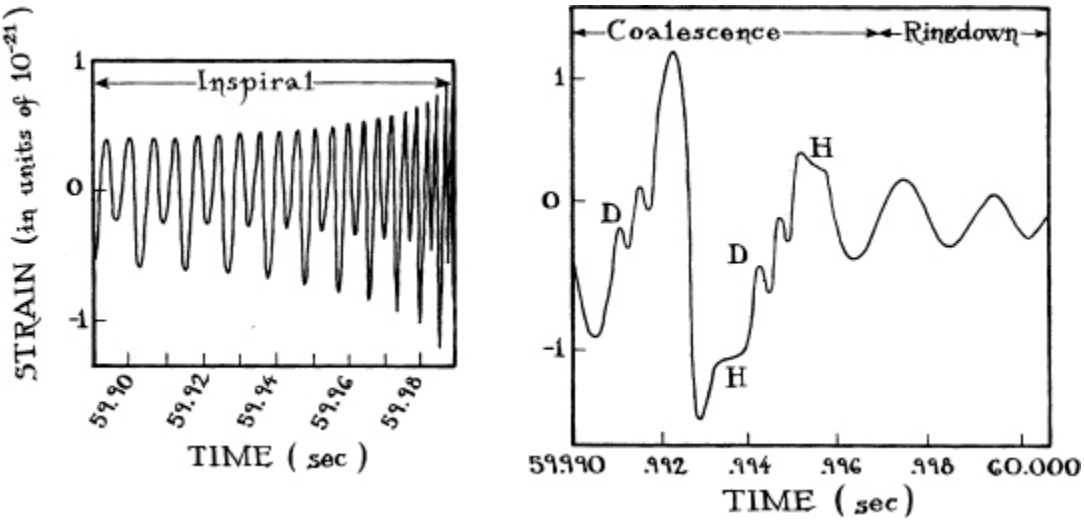


10.9 One of the two waveforms produced by the coalescence of two black holes. The wave is plotted upward in units of 10^{-21} time is plotted horizontally in units of seconds. The first graph shows only the last 0.1 second of the inspiral part of the waveform; the preceding minute of the waveform is similar, with gradually Increasing amplitude and frequency. The second graph shows the last 0.01 second, on a stretched-out scale. The *Inspiral* and *Ringdown* segments of the waveform are well understood, in 1993, from solutions of the Einstein field equation. The coalescence segment is not at all understood (the curve shown is my own fantasy); future supercomputer simulations will attempt to compute it. In the text these simulations are presumed to have been successful in the early twenty-first century.

The waveforms have three segments:

- The minute-long first segment (of which only the last 0.1 second is shown in Figure 10.9) has oscillating strains that gradually grow in amplitude and frequency; these are precisely the waveforms expected from the *inspiral* of two objects in a binary orbit. The fact that alternate waves are smaller and larger indicates that the orbit is somewhat elliptical rather than circular.
- The O.Ol-second-long middle segment matches almost perfectly the waveforms predicted by recent (early twenty-first century) supercomputer simulations of the *coalescence* of two black holes to form one; according to the simulations, the humps marked “H” signal the touching and merging of the holes’ horizons. The double wiggles marked “D,” however, are a new discovery, the first one made by the new specialty interferometers. The older, workhorse interferometers had never been able to detect these wiggles because of their high frequency, and they had never yet been seen in any supercomputer simulations. They are a new challenge for theorists to